

Lab 7 — Create a Floor and a Roof by Footprint

Autodesk Revit Architecture Masterclass (C508) · Topic 02: Part Lists for Modeling in Revit · Tertiary Infotech Academy Pte Ltd

Goal

Create a mezzanine floor by sketching its boundary, edit the sketch with Align and Trim into a closed loop, then complete the envelope with a footprint roof whose slopes you define.

You'll build

A mezzanine floor in the store-room area and a sloped roof closing the building envelope.

Tools: Revit · Floor: Architectural · Roof by Footprint · Shaft Opening

Files in this lab

- ModelCreation.rvt (open in Autodesk Revit)

Step-by-step

1. Open the upper-level plan and start the architectural Floor tool.

Architecture ► Build ► Floor ► Floor: Architectural

2. Pick the bounding walls, then switch to Line to close the open edge.

Modify | Create Floor Boundary ► Pick Walls ► Draw ► Line

3. Use Align and Trim to make the sketch one closed loop, then finish.

Modify ► Align / Trim ► ✓ Finish Edit Mode

4. Cut a stair opening through the new floor with the Shaft tool.

Architecture ► Opening ► Shaft ► sketch the opening ► Finish

5. Create the roof from the building footprint on the roof level.

Architecture ► Build ► Roof ► Roof by Footprint ► Pick Walls

6. Set which boundary lines define a slope, then finish and view in 3D.

Select a line ► Properties ► Defines Roof Slope ✓ ► ✓ Finish

Test it

The mezzanine floor appears with a clean shaft opening, and the 3D view shows a sloped roof sitting on the footprint — no warnings about open loops.

© 2026 Tertiary Infotech Academy Pte Ltd · UEN: 201200696W